Perhaps the most commonly used class of PPGs is that of heterolytic PPGs, featuring the widely applied BODIPY- and coumarin-PPGs.[23–27] The mechanism of uncaging of heterolytic PPGs relies on excited state heterolysis of a σ bond, resulting in formation of a contact ion pair intermediate (CIP) consisting of the PPG-cation and payload-anion (Figure 1a, CIP).[25,26]

In the next step, the components of this transient intermediate either unproductively recombine or diffuse apart, the latter process resulting in productive payload release and usually solvent-capture of the PPG-cation. The nature of the CIP intermediate is crucial in determining the photochemistry of PPGs. For example, engineering contact ion pair stability has been proven to be a powerful strategy for improving the quantum yield of heterolytic PPGs.[25,26,28–30] In principle, both the anionic and cationic constituent of the CIP could be modified to stabilize their charges, resulting in overall CIP stabilization and, possibly, an improved QY. Improvement of the QY through stabilization of the anionic partner (*i.e.* the payload) was first presented by Bendig *et al.* who demonstrated the strong relationship between the ability of the payload of a coumarin PPG to stabilize the incipient negative charge in the CIP (*i.e.*, the pK_a of its conjugated acid, H-PL) and the rate constant of the heterolysis step in the photochemical payload release reaction (**k_1**).[25] Using dimethylaminocoumarin as a model PPG (Figure 1a, **DMACM-PL**) and a variety of payloads with pK_a's of their conjugated acids ranging from −1.5 to 6.5, they observed a strong correlation between these pK_a values and the rate constant **k_1**, with lower pK_a values resulting in higher rate constants. However, engineering the QY of PPGs through anion stabilization involves payload functional group alteration and is undesirable, as it fundamentally changes the chemical nature of the bioactive compound to be released. Therefore, most strategies to improve the QY of heterolytic PPGs focus on stabilizing the cationic member of the CIP (*i.e.* the PPG) or creating new ways for the liberation of constituents from the CIP.[19,28,29,31,32]

While engineering the QY of PPGs through anion stabilization often results in undesired payload functional group alteration, this problem can be circumvented. A strategy through which the leaving group H-PL's pK_a can be lowered without affecting the chemical structure of the ultimately released payload is through using a self-immolative linker that is installed between the actual payload and the PPG. This strategy has been utilized to uncage alcohols and also amines from heterolytic PPGs.[33,34] The pK_a values of alcohols (~16) is often too high to be uncaged efficiently from these PPGs via a direct ether linkage.[35] However, installing alcohol payloads via carbonate linkages onto the α-carbon of a heterolytic PPG allows for the uncaging of alcohols with QYs comparable to that of carboxylic acid payloads.[33,34] After photochemical release of the carbonate, a decarboxylation step follows to ultimately yield the alcohol payload (Figure 1b).

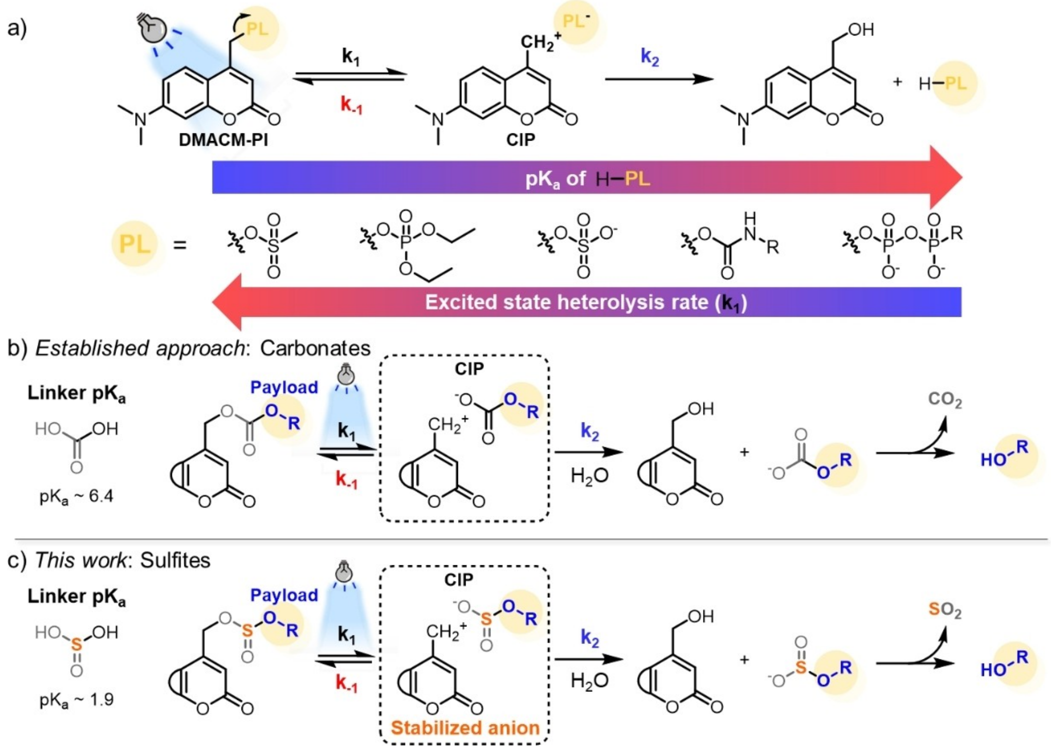

Figure 1. Anion stability and uncaging Quantum Yield. a) The model PPG DMACM studied by Bendig *et al.*[25] and the mechanism of payload release after excitation. A correlation was observed between the pK_a values of a range of structurally different payloads and the rate constant **k_1** of the heterolysis step. b) The current conventional strategy for photocaging alcohol payloads through the use of a self-immolative carbonate linker. c) This work: QY enhancement through engineering the self-immolative linker